

Project Report: Multi-Objective Portfolio Optimization

1 Introduction

This project focuses on optimizing investment portfolios by considering multiple conflicting objectives: maximizing expected return, minimizing risk (variance), and maximizing momentum. Two primary multi-objective optimization techniques were implemented and compared: the Scalarization (Weighted Sum) Method and the Heuristic (NSGA-II) Method.

2 Data Preparation and Feature Engineering

The analysis utilized historical S&P 500 data (`SP500_2011_2025.csv`). The following steps were performed:

- **Data Loading and Cleaning:** The dataset was loaded, column names were standardized, and the 'date' column was converted to datetime objects. Missing values in key columns were dropped.
- **Log Returns:** Daily log returns were calculated as the primary measure of profit.
- **Realized Volatility:** A 21-day rolling standard deviation of log returns, annualized, was used as a measure of risk.
- **Lagged Features:** `log_return` and `realized_volatility` were lagged by one period to create `return_lag_1` and `lagged_volatility`, which serve as features for the current day's decision.
- **Technical Indicators:**
 - **Relative Strength Index (RSI):** Calculated over a 14-period window to capture trend strength.
 - **Moving Average Convergence Divergence (MACD) Difference:** Calculated with standard parameters (12-day EMA, 26-day EMA, 9-day signal line) to indicate momentum.
- **Log Average Volume:** The logarithm of the trading volume was used as a proxy for liquidity.

For the optimization, a subset of 200 data points (sampled every 15th row after dropping NaNs) was selected to represent the asset universe for which weights needed to be determined. The R (returns), M (MACD differences), and `cov_matrix` (derived from lagged volatility) were prepared for the optimization algorithms.

3 Optimization Objectives

The three objectives for portfolio optimization were defined as follows:

- **Maximize Return:** Achieved by maximizing the dot product of asset returns (R) and portfolio weights (x).
- **Minimize Risk:** Achieved by minimizing the portfolio variance, calculated as:

$$x^T \cdot \text{cov_matrix} \cdot x$$

where `cov_matrix` is a diagonal matrix of squared lagged volatilities.

- **Maximize Momentum:** Achieved by maximizing the dot product of the hyperbolic tangent of MACD differences ($\tanh(M)$) and portfolio weights (x). The $\tanh$ function was applied to bound the momentum values.

Constraints for both methods included:

1. All portfolio weights must sum to 1.0 ($\sum x_i = 1$).
2. All portfolio weights must be between 0 and 1 (no short selling, $0 \leq x_i \leq 1$).

4 Scalarization Technique (Normalized Weighted Sum Method)

This method converts the multi-objective problem into a single-objective problem by assigning weights to each objective and summing them. To ensure fair comparison across objectives with different scales, a normalization step was performed:

- **Ideal and Nadir Points:** The best (ideal) and worst (nadir) possible values for each individual objective were found by optimizing each objective independently.
- **Normalization:** Each objective's raw value was normalized using the formula:

$$\frac{\text{Raw} - \text{Ideal}}{\text{Nadir} - \text{Ideal}}$$

This maps all objective values to a $[0, 1]$ range, where 0 is the ideal and 1 is the nadir.

- **Weighted Sum:** A weighted sum of the normalized objectives was minimized. The sweep involved varying weights (w_1, w_2, w_3) for return, risk, and momentum, ensuring their sum is 1.0. This generated a set of Pareto-optimal solutions for various preferences.

5 Heuristic Technique (NSGA-II - Non-Dominated Sorting Genetic Algorithm-II)

NSGA-II is a popular evolutionary algorithm for multi-objective optimization that directly searches for a set of Pareto-optimal solutions without requiring explicit weights. Key components implemented were:

- **Fast Non-Dominated Sorting:** Efficiently sorts the population into different Pareto fronts (non-dominated layers).
- **Crowding Distance Assignment:** Measures the density of solutions in the objective space, ensuring diversity among solutions on the same front.
- **Crowded Binary Tournament Selection:** Selects individuals for the next generation based on rank (preferring lower fronts) and then crowding distance (preferring less crowded solutions).
- **Variation Operators:**
 - **Simulated Binary Crossover (SBX):** Generates two offspring from two parents for real-coded variables.
 - **Polynomial Mutation:** Introduces small perturbations to the offspring's genes.
- **Survivor Selection:** Combines parent and offspring populations, then uses fast non-dominated sorting and crowding distance to select the best `pop_size` individuals for the next generation.

6 Comparison and Metrics

Both methods aim to approximate the Pareto front. Their performance was evaluated using the following metrics:

- **SPACING (Uniformity of Solutions):** Measures how evenly distributed the solutions are along the Pareto front. *Lower values are better.*
 - **NSGA-II:** 0.012979
 - **Weighted Sum:** 0.048055

Note: NSGA-II shows significantly better spacing, indicating a more uniform distribution of solutions.
- **MAXIMUM SPREAD (Coverage):** Measures the extent of the Pareto front found. *Higher values are better.*
 - **NSGA-II:** 0.281323
 - **Weighted Sum:** 0.884832

Note: The Weighted Sum method achieved a much larger maximum spread, covering a broader range of the objective space.

- **GENERATIONAL DISTANCE (Convergence):** Measures the average distance from the obtained Pareto front to a reference or ‘true’ Pareto front (approximated here by the merged non-dominated solutions of both methods). *Lower values are better.*
 - **NSGA-II:** 0.00164181
 - **Weighted Sum:** 0.00000000

Note: The Weighted Sum method shows an extremely low generational distance, suggesting its solutions are very close to the ‘true’ front identified by combining both sets of solutions.
- **Hypervolume (Overall Quality):** A comprehensive metric that measures the volume of the objective space dominated by the Pareto front with respect to a reference point (nadir). *Higher values are better.*
 - **NSGA-II:** 0.125450
 - **Weighted Sum:** 0.142538

Note: The Weighted Sum method produced a higher hypervolume, indicating a slightly better overall quality and coverage of the objective space.

7 Conclusion

The comparison reveals distinct characteristics of each approach:

Weighted Sum Method

- **Strengths:** Excels in *convergence* and *maximum spread*, effectively covering a wide range of the Pareto front and finding solutions very close to the optimal. Its hypervolume is also slightly better. It is simpler to implement for convex Pareto fronts.
- **Weaknesses:** Suffers from poorer *spacing*, meaning the solutions found might be clustered, especially if the weights are swept linearly. It is known to struggle with non-convex Pareto fronts.

NSGA-II Method

- **Strengths:** Demonstrates superior *spacing*, leading to a more diverse and evenly distributed set of solutions. This is crucial for exploring the entire trade-off landscape.
- **Weaknesses:** While still effective, its *maximum spread*, *generational distance*, and *hypervolume* were slightly less optimal compared to the Weighted Sum method in this particular run.

Overall: For this specific problem instance, the **Weighted Sum method** appears to yield solutions that are closer to the ‘true’ Pareto front and cover a wider range of the objective space (higher maximum spread and hypervolume). However, **NSGA-II**

provides a more **uniform distribution of solutions**, which is valuable for decision-makers who need a diverse set of options across the trade-off surface. The choice between the two often depends on the specific shape of the Pareto front and the decision-maker's priorities regarding convergence vs. diversity.